TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Evaluation of Pre-treated Textile Materials

Judging preparation by test rather than by eye: absorbency by drop, sinking and wicking; whiteness assessment; and the chemical spot tests for residual size, chlorine, peroxide and alkali.

TE-2206 · TEXTILE CHEMISTRY & COLORATION-I SESSIONAL · EXPERIMENT 3 · LEVEL 2 TERM 2

1. Aim and theory

Aim: to evaluate textile material prepared in Experiment 2 — cotton after desizing, scouring and bleaching, and the scoured protein and polyester samples — using absorbency tests, whiteness assessment and chemical tests for residues, and to decide from the evidence whether the preparation is fit for dyeing. Preparation faults are cheap to correct before dyeing and ruinous after it: an unlevel dyeing almost always traces back to preparation, so the prepared goods are passed or failed here, not at the dyeing stage.

Absorbency — what is being measured

Absorbency is the practical proof that hydrophobic matter (size, waxes) has left the fibre and that the capillary system of the yarn is open to water. All three tests measure the same property in different geometries:

- **Drop test:** a standard drop of water is placed on the fabric and timed until its glossy reflection disappears — the moment the drop has been taken in.
- **Sinking test:** a small sample is laid on a still water surface and timed until it sinks; trapped air and hydrophobic surfaces keep it afloat.
- **Wicking (capillary-rise) test:** a strip hangs with its lower end in water and the height the water front climbs in a fixed time is measured. Rise is fast at first and slows (the column's weight grows against the capillary pull), so heights must always be quoted with their time.

The residual tests — spot chemistry

Each residue is hunted with a reaction that produces a colour only if the residue is there:

- **Residual size (starch):** iodine solution on the fabric. Iodine slips inside the amylose helix of starch and the complex is an intense **blue-black**. Any strong blue-black stain after desizing means starch is still present. (PVA size answers differently — a brownish-green with iodine–boric acid reagent — which is how the two sizes are told apart.)
- **Residual chlorine** (after a hypochlorite bleach): chlorine left in the goods oxidises iodide to iodine, so starch–iodide paper, or a drop of acidified KI–starch solution on an extract of the sample, turns **blue** if free chlorine remains:

  $$Cl_2 + 2\ I^- \rightarrow I_2 + 2\ Cl^- \quad (I_2 + \text{starch} \rightarrow \text{blue})$$

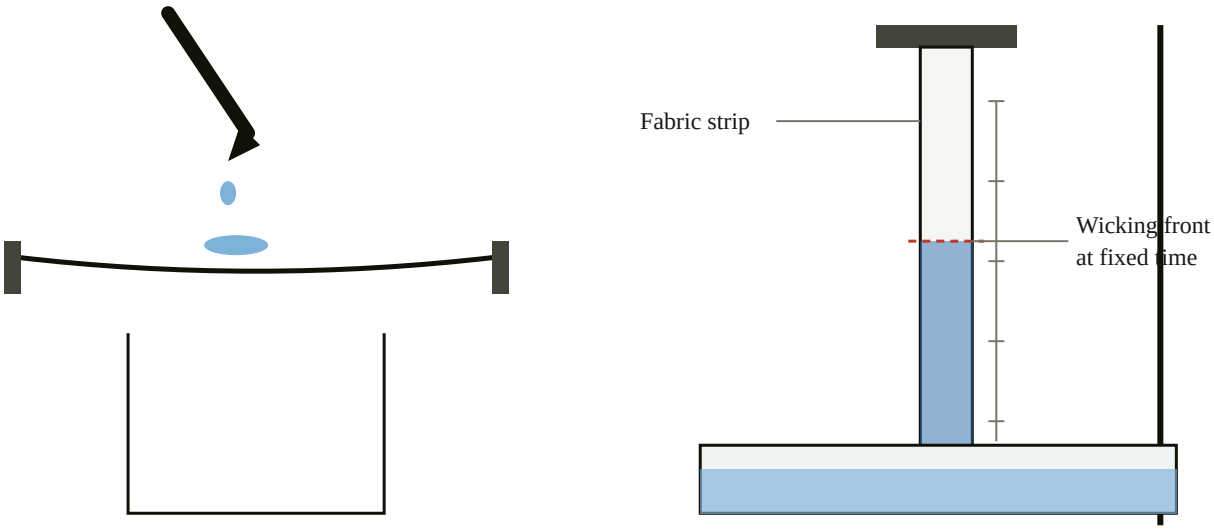
  Residual chlorine must be removed (an antichlor bath, e.g. sodium bisulphite) because it tenders cellulose on storage and destroys dyes in the next bath.
- **Residual peroxide** (after a peroxide bleach): peroxide also oxidises iodide to iodine — blue with KI–starch — and with titanium(IV) sulphate solution gives a **yellow** peroxotitanium colour. Peroxide must test absent (or be killed with a catalase/reducing rinse) before dyeing, or it will bleach the dye as fast as the dye is applied.
- **Residual alkali:** a drop of phenolphthalein solution on scoured goods turns **pink** if alkali remains; thoroughly rinsed material leaves it colourless. A pH paper pressed on the wet sample does the same job numerically.

Whiteness

Whiteness is judged either **visually** — samples compared side by side under standard daylight (D65 illuminant), ranked against a set of whiteness standards — or **instrumentally**, where a spectrophotometer computes a whiteness index (e.g. the CIE whiteness formula) from the reflectance spectrum. The instrument removes the observer, but not the physics: a sample can gain whiteness by bleaching (colour destroyed) or by optical brightening (blue fluorescence added), and only the reflectance curve tells which. For this experiment, report the method with the result — "whiter" without a method is an opinion.

2. Apparatus and procedure

- Prepared samples from Experiment 2 (grey sample retained for comparison), stopwatch, pipette or dropper delivering a standard drop (~0.05 mL), beaker of distilled water, clamp stand and ruler, a trough with water containing a little direct dye (the dye makes the wicking front visible), iodine solution, starch–iodide paper/solution, titanium sulphate solution (or acidified KI–starch), phenolphthalein solution, pH paper, white tile, standard light box if available.



A — Drop test: time until the drop's shine vanishes

B — Wicking test: dyed water climbs the strip

Fig. 1 — The two absorbency geometries of this experiment: (A) the drop test on fabric held flat, (B) vertical wicking, read against the scale at a stated time.

Procedure

1. **Drop test:** lay the conditioned sample flat and taut. From a fixed small height, place one standard drop and start the watch; stop it when the drop's bright reflection just disappears. Repeat at five places and average. Run the grey sample alongside.
2. **Sinking test:** cut 2 cm × 2 cm pieces, lay one gently on a still distilled-water surface and time its sinking. Repeat three times.
3. **Wicking test:** clamp a 2.5 cm wide strip vertically, lower end just dipping into the dyed water. Read the front height at 5, 10 and 30 minutes.
4. **Whiteness:** compare the bleached sample with the scoured-only and grey samples under the standard light, rank them, and (if an instrument is available) record the whiteness index of each.
5. **Spot tests** on small cuttings or on a hot-water extract of the sample: iodine for size; starch–iodide for chlorine (on hypochlorite-bleached material) and KI–starch or titanium sulphate for peroxide (on peroxide-bleached material); phenolphthalein for alkali. Record each as absent / trace / strong, with a distilled-water blank for comparison.

Guide values (teaching-laboratory guides for cotton, not universal specifications)

TEST	GREY COTTON	WELL-PREPARED COTTON
Drop test	Drop stands, > 60 s	≤ 5 s, typically 1–3 s
Sinking test	Floats for minutes	Sinks within a few seconds

Wicking height, 10 min	Almost nil	≥ 8 cm
Residual size (iodine)	Strong blue-black	No more than a faint stain
Residual chlorine / peroxide	—	Must test absent before dyeing
Residual alkali (phenolphthalein)	—	No pink

3. Worked example — judging a batch

A peroxide-bleached cotton lot from Experiment 2 gave the results below. The last column is the reading this experiment exists to teach.

TEST	RESULT	INTERPRETATION
Drop test	15 s	Too slow (guide ≤ 5 s): hydrophobic matter remains
Wicking, 10 min	3.1 cm (average rate 0.31 cm/min)	Poor (guide ≥ 8 cm): agrees with the drop test
Residual size	Distinct blue-black	Starch size still present → desizing was incomplete
Whiteness	Good, matches the passed reference lot	Bleach stage itself worked
Residual peroxide	Positive (KI–starch blue)	Peroxide still in the goods
Residual alkali	No pink	Rinsing after scouring was adequate

VERDICT

Do not dye this lot. The absorbency tests and the iodine test tell one consistent story — the desize (and with it the scour) did not finish, so wax and size still block the capillaries; dyeing now would come up unlevel and pale, and the fault would be blamed on the dye. The peroxide finding is a separate, simpler fault: an extra rinse (or a catalase treatment) removes it. Corrective route: return the lot to desizing and re-scour, rinse out the peroxide, then re-test absorbency and size before releasing it to dyeing. Note the diagnostic pattern — whiteness passed while absorbency failed: bleaching can whiten the surface of a badly scoured cloth, so whiteness alone never passes a lot.

4. Precautions

- Condition all samples alike and test at several places: absorbency varies across a piece, and a single drop is an anecdote, not a result.
- Keep the drop volume and dropping height fixed, and read the wicking front at the exact stated times — the tests are comparative, and sloppy geometry destroys the comparison.
- Run a distilled-water blank beside every spot test. Old iodine solution and contaminated glassware both manufacture false positives.
- Test residues on an extract as well as on the cloth when the result is doubtful; colour on the fabric itself can be masked by the fabric's own shade.
- Judge whiteness under the standard light only. Ordinary room light flatters yellow samples and quarrels between observers are usually quarrels between lamps.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why are three different absorbency tests used when all three measure “wetting”?
2. A wicking strip rises 6.3 cm in 10 minutes. Give the average rate and a pass/fail judgement against this sheet's guide.
3. Write the reaction by which residual chlorine betrays itself to starch–iodide, and state why the residue is dangerous if left.
4. Both chlorine and peroxide give a blue colour with KI–starch. How does the titanium sulphate test separate them?
5. A lot is very white but fails the drop test. Explain how that can happen and what it means for dyeing.
6. Why must residual peroxide be absent before dyeing, while residual whiteness obviously may be present? Frame the answer in what each does to dye.

Answer key

1. They interrogate different geometries and time-scales: the drop test is local surface wetting, the sinking test exposes trapped air and overall hydrophobicity of a small piece, and wicking measures transport along the yarn capillaries — the path dye liquor actually travels. Agreement between them is evidence; any one alone can mislead.
2. Average rate = 6.3/10 = **0.63 cm/min**. Height at 10 min is 6.3 cm, below the ≥ 8 cm guide → **fail**: preparation is incomplete even though the rate sounds respectable; the guide is on the height at the stated time.
3. $Cl_2 + 2\ I^- \rightarrow I_2 + 2\ Cl^-$; the liberated iodine forms the blue starch complex. Left in the goods, chlorine tenders cellulose in storage (oxycellulose formation) and attacks dyes in the following bath.

4. Titanium(IV) sulphate gives a yellow peroxotitanium colour specifically with hydrogen peroxide; chlorine gives no such colour. A yellow with titanium sulphate (plus the KI–starch blue) identifies peroxide; KI–starch blue with a negative titanium test points to chlorine.
5. Bleaching destroys surface colour bodies and can whiten a cloth whose capillaries are still blocked by wax and size that the scour failed to remove. Whiteness judges colour; the drop test judges accessibility. Dyeing needs accessibility — so this lot dyes unlevel despite looking bright.
6. Peroxide is an oxidising agent loose in the bath: it oxidises and destroys dye molecules as they are applied, giving pale, unlevel results — an active chemical interference. Whiteness is a passive optical state of the fibre; it does nothing to the dye at all.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib